# NeuralGraph: complete results of the retrieval research campaign

September 5 to 6, 2026. All runs on LoCoMo (10 conversations, 1,540 questions in four categories: single_hop 282, multi_hop 841, temporal 321, open_domain 96). Models: google/gemma-4-e4b for reranking, answering, and the campaign judge; nomic-embed-text v1.5 for embeddings; qwen/qwen3.6-35b-a3b as an independent judge. All numbers are leakage-free unless marked "old" (the December 2025 run used the gold answer as an acceptance gate and the gold category label for routing; both removed before any run below).

Two metrics are used throughout:

- **Accuracy**: an LLM judge labels the generated answer correct or wrong against the gold answer.
- **Recall@k**: any part of the gold answer (split on commas and "and", parts longer than two characters) appears as a substring in one of the top k retrieved messages. Retrieval-only, no LLM calls. Meaningless for the temporal category, whose gold answers are dates that live in metadata, so temporal is omitted from retrieval tables.

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## 1. Headline: end-to-end accuracy of the shipped system

Shipped configuration: `RETRIEVAL_MODE=local_pairs` (per-agent memory routing, Tesseract top-50, 30 pair-node back-fills) plus open-domain prompt flags, 80-candidate pointwise rerank, 15 memories in the answer prompt (30 for list questions).

### 1.1 Conversations 1 to 5, 744 of 1,540 questions (run stopped early at the user's request)

| Category | n | New system (Gemma, lenient) | Old leaky run, same questions (GPT-4o, lenient) |
|---|---|---|---|
| single_hop | 142 | 73.9% | |
| multi_hop | 400 | 72.8% | |
| temporal | 156 | 73.7% | |
| open_domain | 46 | 56.5% | |
| **overall** | **744** | **72.2%** | **66.9%** |

The two columns use different judges, so the 5.3-point gap is indicative only.

### 1.2 Conversations 1 to 4, 584 questions, under four judges

| Category | n | Old leaky (GPT-4o lenient) | Gemma lenient | Qwen lenient | Qwen strict | Substring |
|---|---|---|---|---|---|---|
| single_hop | 111 | 47.7 | 75.7 | 62.2 | 33.3 | 21.6 |
| multi_hop | 311 | 75.2 | 72.3 | 66.6 | 41.8 | 27.0 |
| temporal | 130 | 71.5 | 78.5 | 71.5 | 60.8 | 36.2 |
| open_domain | 32 | 53.1 | 62.5 | 43.8 | 6.2 | 3.1 |
| **overall** | **584** | **68.0** | **73.8** | **65.6** | **42.5** | **26.7** |

### 1.3 The clean comparison: single_hop, original flat retrieval vs shipped stack

Same 282 questions, same code base, same judge, only retrieval changed.

| Retrieval | Accuracy | recall@10 | recall@50 | seconds per question |
|---|---|---|---|---|
| flat (original Tesseract) | 64.9% | 39.4 | 61.7 | 6.4 |
| hybrid (speaker boost + embedding back-fill) | 67.0% | 44.7 | 61.7 | 8.2 |
| graph (time chain, same-speaker edges, dialogue links appended) | 66.7% | 39.4 | 61.7 | 7.9 |
| **local_pairs (per-agent routing + pair back-fill)** | **73.8%** | 46.8 | 62.8 | 7.9 |

Same 111 single_hop ids from conversations 1 to 4, flat vs shipped, per judge:

| Judge | Flat | Shipped | Gain | Wins / losses |
|---|---|---|---|---|
| Gemma lenient | 66.7 | 75.7 | +9.0 | 32 / 7, p < .001 |
| Qwen lenient | 53.2 | 62.2 | +9.0 | 45 / 13, p < .001 |
| Qwen strict | 22.5 | 33.3 | +10.8 | 38 / 11, p < .001 |
| Substring | 18.0 | 21.6 | +3.6 | 18 / 5, p = .01 |

Where the flat system loses: when the gold text reached the final 15-memory context, accuracy was 77.8%; when it did not (124 of 282 questions), 48.4%.

### 1.4 Full benchmark without the LLM reranker (all 1,540 questions, Ali's ablation)

Same retrieval ( `local_pairs` ) and flags, `RERANK=none` : the answer prompt takes the retriever's native top 15 with no LLM reranking.

| Category | n | No reranker, Gemma lenient | Reranked stack, same question ids (1,112 available) |
|---|---|---|---|
| single_hop | 282 | 65.6% | |
| multi_hop | 841 | 65.0% | |
| temporal | 321 | 53.3% | |
| open_domain | 96 | 52.1% | |
| **overall** | **1,540** | **61.9%** | **68.6% vs 62.2% on the same 1,112** |

Latency: 0.44 s per question end to end (retrieve 0.09 s, answer 0.34 s) versus 8.2 s with the 80-call reranker. So the trade is about 6.5 accuracy points for an 18x speedup. Temporal questions lose the most, which means the reranker's contribution is ordering, and ordering matters most when several dated memories compete. Note the 30 pair back-fill candidates never reach the prompt without a reranker (they sit at positions 51 to 80), so a cheap non-LLM fusion of Tesseract rank and pair-node cosine rank is the obvious next step before shipping the fast path.

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## 2. Retrieval experiments (recall, no LLM calls)

### 2.1 Baseline retrievers, all 1,540 questions

| Retriever | single_hop @10 / @50 | multi_hop @10 / @50 | open_domain @10 / @50 | ALL @10 / @50 |
|---|---|---|---|---|
| pure embedding | 25.9 / 52.1 | 35.1 / 54.8 | 13.5 / 19.8 | 25.3 / 41.9 |
| Tesseract flat | 39.4 / 61.7 | 46.8 / 61.4 | 13.5 / 20.8 | 34.9 / 48.0 |
| Tesseract + graph neighbours mixed into top-50 | 27.3 / 38.3 | 38.3 / 43.5 | 7.3 / 10.4 | 27.5 / 32.9 |

| Retriever | single_hop @10 / @50 | multi_hop @10 / @50 | open_domain @10 / @50 | ALL @10 / @50 |
|---|---|---|---|---|
| Tesseract + graph neighbours appended (recall@all) | 39.4 / 61.7 / 64.5 | 46.8 / 61.4 / 62.2 | 13.5 / 20.8 / 21.9 | 34.9 / 48.0 / 49.0 |
| Tesseract + speaker boost | 44.7 / 61.7 | 49.2 / 61.4 | 13.5 / 20.8 | 37.2 / 48.0 |
| Tesseract + embedding back-fill (recall@all) | 39.4 / 61.7 / 65.2 | 46.8 / 61.4 / 65.6 | 13.5 / 20.8 / 22.9 | 34.9 / 48.0 / 65.5 (3 cats) |

Budget-matched conclusion: 30 extra candidates from graph neighbours add +1.4 recall@all; 30 from plain embedding add +4.1.

### 2.2 Pair nodes (H3): index each reply together with the message before it

| Variant | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain | ALL @10 / @50 / @all |
|---|---|---|---|---|
| pure embedding | 25.9 / 52.1 / 52.1 | 35.1 / 54.8 / 54.8 | 13.5 / 19.8 / 19.8 | 31.3 / 51.4 / 51.4 |
| embedding + pairs | 41.1 / 59.6 / 59.6 | 54.1 / 65.2 / 65.2 | 12.5 / 19.8 / 19.8 | 47.8 / 60.3 / 60.3 |
| Tesseract + embedding back-fill (control) | 39.4 / 61.7 / 65.2 | 46.8 / 61.4 / 65.6 | 13.5 / 20.8 / 22.9 | 42.5 / 58.2 / 62.2 |
| Tesseract + pair back-fill | 39.4 / 61.7 / 68.4 | 46.8 / 61.4 / 69.1 | 13.5 / 20.8 / 24.0 | 42.5 / 58.2 / 65.4 |
| window of 3 messages | 39.4 / 61.7 / 68.1 | 46.8 / 61.4 / 69.6 | 13.5 / 20.8 / 20.8 | 42.5 / 58.2 / 65.4 |

Verdict: works. +3.2 recall@all over the control at the same 30-candidate budget. Window of 3 adds nothing. Extra cost: 5,872 embeddings once.

### 2.3 Scoring-formula fixes (H4, H6)

| Configuration | single_hop @10 / @50 | multi_hop @10 / @50 | open_domain @10 / @50 |
|---|---|---|---|
| Tesseract baseline | 39.4 / 61.7 | 46.8 / 61.4 | 13.5 / 20.8 |
| H4 keyword denominator fix (morphology + thesaurus) | 39.0 / 60.3 | 47.1 / 60.0 | 13.5 / 21.9 |
| H4 morphology only | 39.0 / 61.3 | 47.9 / 60.6 | 11.5 / 20.8 |
| H6a soft temporal penalties | 39.4 / 61.7 | 46.8 / 61.5 | 13.5 / 20.8 |
| H6b no minimum-charge cutoff | 39.4 / 61.3 | 46.8 / 61.2 | 12.5 / 20.8 |
| H6c embedding rescue | 36.5 / 61.0 | 48.2 / 62.8 | 14.6 / 20.8 |
| H4 + H6 + rescue | 36.5 / 59.9 | 47.9 / 62.0 | 13.5 / 20.8 |

Verdict: null. The four scoring stores overlap so heavily that per-store changes reorder nothing. Surprise: the temporal store dominates the fused top-10 for every category because fusion weights are near-uniform.

### 2.4 Two-agent memory split (H9a): treat each speaker as an agent with its own store

| Variant | single_hop @10 / @50 | multi_hop @10 / @50 | open_domain @10 / @50 |
|---|---|---|---|
| Tesseract, pooled (control) | 39.4 / 61.7 | 46.8 / 61.4 | 13.5 / 20.8 |

| Variant | single_hop @10 / @50 | multi_hop @10 / @50 | open_domain @10 / @50 |
|---|---|---|---|
| Tesseract, routed to the named agent's store | 46.8 / 62.8 | 50.3 / 63.3 | 15.6 / 19.8 |
| Tesseract, oracle routing (gold speaker) | 46.5 / 61.7 | 51.0 / 64.0 | 15.6 / 19.8 |
| Tesseract, federated by charge | 36.9 / 60.3 | 42.4 / 59.9 | 11.5 / 20.8 |
| embedding, pooled | 25.9 / 52.1 | 35.1 / 54.8 | 13.5 / 19.8 |
| embedding, routed to the named agent | 46.8 / 63.8 | 52.8 / 64.4 | 13.5 / 17.7 |

The gold-bearing message is spoken by the other agent only 2.9% of the time (0% for single_hop), so name routing equals oracle routing. Federating by charge is worse than pooling because per-store charges are not comparable.

### 2.5 Speaker-swap probe (H9b): rewrite the question with the other speaker's name, 244 single_hop questions

| Retriever | Original recall@10 | Swapped recall@10 | Gap (identity sensitivity) | Still retrieves original gold in top 10 after swap |
|---|---|---|---|---|
| pure embedding | 26.6 | 34.8 | -8.2 | 78.5% |
| Tesseract | 42.6 | 32.0 | +10.7 | 57.7% |
| Tesseract + speaker boost | 48.8 | 26.2 | +22.5 | 45.4% |

Swapping the name raises pure-embedding recall. Measured cause: a speaker's own name appears in 0.1% of their messages, the other speaker's name in 33.4%. A name in the question is an embedding-level pointer to the wrong speaker.

### 2.6 Combined stacks with local routing (H9 follow-up), 1,219 questions (three categories)

| Variant | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain @10 / @50 / @all |
|---|---|---|---|
| embedding, local | 46.8 / 63.8 / 63.8 | 52.8 / 64.4 / 64.4 | 13.5 / 17.7 / 17.7 |
| embedding + pairs, local | 41.1 / 59.2 / 59.2 | 53.7 / 64.0 / 64.0 | 14.6 / 20.8 / 20.8 |
| Tesseract, local | 46.8 / 62.8 / 62.8 | 50.3 / 63.3 / 63.3 | 15.6 / 19.8 / 19.8 |
| **Tesseract local + pair back-fill (shipped)** | 46.8 / 62.8 / **69.5** | 50.3 / 63.3 / **68.4** | 15.6 / 19.8 / **24.0** |
| Tesseract local + plain embedding back-fill (control) | 46.8 / 62.8 / 66.0 | 50.3 / 63.3 / 65.4 | 15.6 / 19.8 / 19.8 |
| embedding local, top 80 (control) | 46.8 / 63.8 / 66.7 | 52.8 / 64.4 / 66.0 | 13.5 / 17.7 / 21.9 |

Pair nodes are a good back-fill and a bad base: inside the top-50 they cost single_hop 5.7 points at recall@10 because expanding a pair pulls in the other agent's message.

### 2.7 Reply-only pair mapping (N3)

| Variant | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain @all |
|---|---|---|---|
| shipped (both members) | 46.8 / 62.8 / 69.5 | 50.3 / 63.3 / 68.4 | 24.0 |
| reply-only back-fill | 46.8 / 62.8 / 68.4 | 50.3 / 63.3 / 67.1 | 21.9 |

| Variant | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain @all |
|---|---|---|---|
| embedding + pairs reply-only, as base | 45.7 / 62.1 / 62.1 | 55.4 / 64.2 / 64.2 | 18.8 |
| pairs only, reply-only | 45.4 / 62.1 / 62.1 | 55.4 / 64.2 / 64.2 | 18.8 |

Verdict: negative. Of 23 questions lost, 14 had the gold only in the other agent's message; under local routing the both-member pair link is the only channel that lets it in.

### 2.8 Mega search (M1): vector + BM25 + entity-graph PageRank hop, fused by reciprocal rank, 1,219 questions

| Retriever | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain @10 / @50 / @all | ALL @10 / @50 / @all |
|---|---|---|---|---|
| shipped (Tesseract local + pairs) | 46.8 / 62.8 / 69.5 | 50.3 / 63.3 / 68.4 | 15.6 / 19.8 / 24.0 | 46.8 / 59.7 / 65.1 |
| vector + BM25 | 46.8 / 62.1 / 65.2 | 55.2 / 66.0 / 67.4 | 12.5 / 18.8 / 19.8 | 49.9 / 61.4 / 63.2 |
| vector + graph hop | 47.2 / 64.9 / 67.4 | 57.8 / 65.3 / 66.8 | 14.6 / 20.8 / 22.9 | 51.9 / 61.7 / 63.5 |
| vector + BM25 + graph | 47.5 / 62.4 / 67.0 | 55.2 / 64.6 / 67.3 | 12.5 / 21.9 / 22.9 | 50.0 / 60.7 / 63.7 |
| vector + graph + Tesseract, then pairs (regex entities) | 51.1 / 65.6 / 68.1 | 59.1 / 66.6 / 68.5 | 14.6 / 21.9 / 24.0 | 53.7 / 62.8 / 64.9 |
| same, LLM-built entity graph (M2) | 49.6 / 64.5 / 67.7 | 56.7 / 66.7 / 68.6 | 14.6 / 19.8 / 24.0 | 51.8 / 62.5 / 64.9 |
| same, LLM + regex entities | 50.7 / 64.9 / 68.4 | 58.9 / 66.7 / 68.5 | 14.6 / 20.8 / 24.0 | 53.5 / 62.7 / 65.0 |
| LLM graph channel alone | 18.1 / 25.9 / 29.4 | 24.9 / 33.2 / 36.0 | 5.2 / 8.3 / 8.3 | 21.7 / 29.5 / 32.3 |

BM25 adds nothing once vectors and pairs exist. The graph hop is the only graph mechanism that beat plain embedding at the retrieval level (+6.9 recall@10 overall, +8.8 on multi_hop). See section 3.7 for why it still loses end to end.

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## 3. End-to-end experiments (accuracy, same 100 single_hop ids unless noted, shipped retrieval, Gemma lenient judge)

Control for every row: 78 of 100. Gold text present in the final 15-memory context: 71%. Accuracy given gold present: 88.7%.

### 3.1 Listwise reranker (H1)

| Reranker | Accuracy | Rerank ms | End-to-end ms | Gold in context | Accuracy given gold | LLM calls per question |
|---|---|---|---|---|---|---|
| pointwise 0 to 3, one call per candidate (control) | 78 | 7,391 | 7,948 | 71% | 88.7% | 80 |
| listwise, 8 batches of 10, 0 to 10 | 74 | 3,258 | 3,816 | 71% | 84.5% | 8 |
| listwise + second global pass | 72 | 5,845 | 6,387 | 71% | 83.1% | 9 |

Verdict: latency win, accuracy loss. Gold-in-context is identical across all three: the reranker only reorders, retrieval decides.

### 3.2 Answer context size (H2, N1)

| Memories in prompt (plain / list questions) | Accuracy | Gold in context | Accuracy given gold | Answer ms |
|---|---|---|---|---|
| 5 / 10 | 69 | 57% | 87.7% | 180 |
| 10 / 10 | 71 | 60% | 88.3% | 190 |
| 10 / 15 | 71 | 65% | 83.1% | 250 |
| **15 / 30 (default)** | **78** | 71% | 88.7% | 479 |
| 30 / 30 | 76 | 71% | 87.3% | 506 |
| 50 / 50 | 69 | 73% | 78.1% | 593 |

Post-rerank rank of the gold message: 1 to 5 for 52 questions, 6 to 15 for 15, 16 to 30 for 4, absent for 29. Widening adds no gold and distracts the model; narrowing loses corroborating second mentions. The default is the optimum.

### 3.3 LLM router vs regex router (H7)

| Category | Regex (control) | LLM router | Agreement | Added latency |
|---|---|---|---|---|
| single_hop | 78 / 100 | 79 / 100 | 52% | +105 ms |
| temporal | 78 / 100 | 76 / 100 | 97% | +119 ms |

Verdict: null. Regex already sends 97 of 100 temporal questions to the temporal mode. The two routers disagree on half of single_hop yet score the same.

### 3.4 Open-domain prompt flags (H5), all 96 open_domain questions

| Configuration | Accuracy | Abstentions |
|---|---|---|
| control | 41 / 96 = 42.7% | 27 |
| keep context in world mode + hedge routing | 43 / 96 = 44.8% | |
| + memories-plus-world-knowledge prompt for inference mode | 55 / 96 = 57.3% | 13 |

single_hop with the flags on: 77 vs 78 (no regression). Under the strict independent judge the gain disappears (see 4.2); about 4 of the 17 wins are judge leniency on hedged answers.

### 3.5 Pair-aware reranker and answer prompt (N4)

| Configuration | Accuracy | Rerank ms | Gold in context | Accuracy given gold |
|---|---|---|---|---|
| control | 78 | 7,349 | 71% | 88.7% |
| reranker sees previous message | 71 | 8,479 | 69% | 82.6% |
| answer prompt shows previous message | 75 | 7,989 | 73% | 83.6% |
| both | 66 | 8,197 | 70% | 78.6% |

Verdict: negative. The previous line dilutes the candidate rather than explaining it. Pair context belongs in retrieval only.

### 3.6 Answer model swap (H8): Gemma-4-e4b vs Qwen3.6-35b-a3b for the answer call only

| | Gemma answers | Qwen answers |
|---|---|---|
| accuracy, Gemma lenient judge | 78 | 71 |
| accuracy, Qwen lenient judge | 64 | 65 |
| accuracy, Qwen strict judge | 38 | 42 |
| substring | 24 | 29 |
| gold in context | 71 | 71 |
| "Not found" answers with gold present | 4 | 11 |
| answer latency ms | 501 | 913 |

Verdict: a judge artifact. The sign flips once the judge is not the answering model's family.

### 3.7 Mega search and LLM-built graph end to end (M1, M2)

| Category, same 100 ids | Shipped | Regex-entity mega search | LLM-entity mega search |
|---|---|---|---|
| single_hop | 78 | 68 (3 wins / 13 losses) | 70 (4 / 12) |
| multi_hop | 75 | 76 | 77 (8 / 6) |

Accuracy given gold in context on single_hop: 88.6 shipped, 79.1 regex graph, 82.6 LLM graph. The entity hop fills the candidate window with topically related non-answers that the reranker and answer model prefer. LLM extraction (138 to 354 resolved entities and 430 to 712 triples per conversation, versus about 2,000 regex entities) recovers 2 points and does not fix it.

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## 4. Judge sensitivity (N5)

### 4.1 Same answer files, four graders

| Answer file | n | Gemma lenient (campaign) | Qwen lenient | Qwen strict | Substring |
|---|---|---|---|---|---|
| flat, single_hop | 282 | 64.9 | 51.4 | 21.6 | 13.5 |
| shipped, single_hop | 282 | 73.8 | 62.8 | 31.2 | 18.1 |
| H8 Gemma answers | 100 | 78.0 | 64.0 | 38.0 | 24.0 |
| H8 Qwen answers | 100 | 71.0 | 65.0 | 42.0 | 29.0 |
| H5 control, open_domain | 96 | 42.7 | 27.1 | 11.5 | 6.2 |
| H5 treatment, open_domain | 96 | 57.3 | 39.6 | 10.4 | 9.4 |

### 4.2 Does the verdict survive the judge?

| Pair | Gemma lenient | Qwen lenient | Qwen strict | Substring | Verdict |
|---|---|---|---|---|---|
| flat to shipped | +8.9 | +11.3 | +9.6 | +4.6 | stable, works under all |
| Gemma to Qwen answers (H8) | -7.0 | +1.0 | +4.0 | +5.0 | sign flips, null |
| open-domain flags (H5) | +14.6 | +12.5 | -1.0 | +3.1 | vanishes under strict, null |

Agreement with the campaign judge: Qwen lenient 86.5% (kappa 0.72), Qwen strict 59.4% (kappa 0.30), substring 49.1% (kappa 0.17). Gemma-correct answers average 53 characters, wrong ones 21; the strict judge shows no length bias. Judge model alone moves scores about 12 points with an

identical prompt; prompt leniency about 30 points, almost all on multi-item gold answers (66% of single_hop golds).

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## 5. Summary table of every experiment

| # | Experiment | Level | Result | Verdict |
|---|---|---|---|---|
| 0 | Leakage removal | harness | scores now honest | required |
| H9 | Per-agent memory routing | retrieval + e2e | recall@10 +7.4; e2e +8.9 with pairs | **works** |
| H3 | Pair nodes as back-fill | retrieval | +3.2 recall@all | **works** |
| H5 | Open-domain prompt flags | e2e | +14.6 lenient, -1.0 strict | lenient-judge only |
| hybrid | Speaker boost + embedding back-fill | e2e | +2.1 | small |
| graph | Graph-neighbour expansion | retrieval + e2e | +1.4 recall, +1.8 e2e | no better than embedding back-fill |
| H1 | Listwise reranker | e2e | -4, latency -56% | speed only |
| H2 | Wider context (30, 50) | e2e | -2, -9 | hurts |
| N1 | Narrower context (10, 5) | e2e | -7, -9 | hurts |
| H7 | LLM router | e2e | +1 / -2 | null |
| H4 | Keyword formula fix | retrieval | -0.4 to -1.4 | null |
| H6 | Softer temporal penalties | retrieval | within 0.2 | null |
| N3 | Reply-only pairs | retrieval | -1 to -2 | hurts |
| N4 | Pair-aware reranker / prompt | e2e | -7 / -3 / -12 | hurts |
| H8 | 35B answer model | e2e | -7 lenient, +4 strict | judge artifact |
| M1 | Mega search (vector + BM25 + graph hop) | retrieval + e2e | +6.9 recall@10; e2e -10 single_hop, +1 multi_hop | hurts |
| M2 | LLM-built entity graph in mega search | retrieval + e2e | +5.0 recall@10; e2e -8 single_hop, +2 multi_hop | hurts |
| N5 | Judge sensitivity | evaluation | 51-point spread by grader | paper-grade |

Files: per-question results in `demo/results/*.json`; per-experiment reports in `docs/research/`; rescoring script `docs/research/n5_rescore.py`.

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## 6. Latency per stage (seconds per question, mean; last column p95 end to end)

Measured on the same Mac and LM Studio server (gemma-4-e4b) unless marked old. Rerank = one model call per candidate (50 for flat, 80 for the shipped stack). Judge time is excluded.

| Run | n | retrieve | rerank | answer | end-to-end mean | end-to-end p95 |
|---|---|---|---|---|---|---|
| flat retrieval (original), single_hop | 282 | 0.14 | 5.60 | 0.68 | 6.43 | 7.44 |
| hybrid, single_hop | 282 | 0.13 | 7.51 | 0.55 | 8.20 | 9.36 |
| graph neighbours appended, single_hop | 282 | 0.13 | 7.37 | 0.44 | 7.94 | 8.25 |
| shipped local_pairs, single_hop | 282 | 0.08 | 7.34 | 0.46 | 7.89 | 8.24 |
| shipped, conversations 1-4, all categories | 590 | 0.08 | 7.68 | 0.39 | 8.17 | 10.99 |

| Run | n | retrieve | rerank | answer | end-to-end mean | end-to-end p95 |
|---|---|---|---|---|---|---|
| shipped, 100 single_hop (control) | 100 | 0.07 | 7.39 | 0.48 | 7.95 | 8.27 |
| Qwen-35B answer model, 100 single_hop | 100 | 0.08 | 7.28 | 0.89 | 8.26 | 8.88 |
| open-domain flags, 96 open_domain | 96 | 0.09 | 7.39 | 0.32 | 7.83 | 8.24 |
| mega search (regex graph), 100 single_hop | 100 | 0.10 | 7.48 | 0.51 | 8.10 | 8.51 |
| mega search (LLM graph), 100 single_hop | 100 | 0.09 | 7.28 | 0.48 | 7.86 | 8.20 |
| mega search (regex graph), 100 multi_hop | 100 | 0.08 | 7.39 | 0.46 | 7.93 | 8.31 |
| 30-memory context, 100 single_hop (H2_treatment30.json) | 100 | 0.07 | 7.36 | 0.51 | 7.95 | 8.25 |
| shipped, NO reranker, all 1,540 | 1540 | 0.09 | 0.00 | 0.34 | 0.44 | 0.9 |
| old Dec-2025 run (Ollama, qwen2.5-7b, leaky) | 1540 | 0.33 | 1.49 | 0.39 | 2.41 | 3.15 |

Reading: retrieval is 80 to 140 ms and gets faster with per-agent stores; the shipped stack costs about 2 s more per question than flat because it reranks 80 candidates instead of 50, which is the price of the +9 accuracy. The listwise reranker halves rerank time at a 4-point accuracy cost. Old-run rerank latency reflects a different server (Ollama) and model, not the architecture.